

Stress Testing Unlearning Algorithms

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Abstract

Recently, machine unlearning, the removal of specific training data influence from a model, has gained increasing attention. In large language models (LLMs), unlearning is particularly challenging due to the ambiguity of inputs and outputs. Consequently, rigorous evaluation is critical for assessing both safety and utility, and for driving progress in unlearning methods. We identify two key shortcomings in existing unlearning benchmarks: (1) they do not actively test whether unlearned information can still be forcibly extracted, and (2) they fail to evaluate performance preservation on boundary questions, benign queries that are semantically close to the unlearned content. Here we introduce WMDP++, an extension of WMDP that addresses these gaps by incorporating targeted extraction of unlearned information and systematic evaluation on boundary questions. WMDP++ provides a more stringent and informative benchmark for evaluating unlearning in LLMs.

Introduction

Modern foundational models are trained on vast, web-scale datasets comprising billions of data points that cannot be thoroughly screened or curated prior to training (Brown et al. 2020). As a result, these datasets inevitably absorb undesirable content, including private or personally identifiable information, copyrighted material, and hazardous knowledge such as instructions for synthesizing dangerous substances or exploiting cybersecurity vulnerabilities. Such findings underscore the urgent need to remove the influence of specific training data from already-deployed models. However, retraining these models from scratch after the problematic data has been identified is often prohibitively expensive, as the computational cost of a single training run for a large-scale model can reach hundreds of millions of dollars.

Machine unlearning has emerged as a promising solution to this challenge, aiming to surgically erase the influence of targeted training data from a model without incurring the cost of full retraining (Liu et al. 2025; Ren et al. 2025). A variety of methods have been proposed, ranging from optimization-based approaches such as Gradient Ascent (GA) and Negative Preference Optimization (NPO) (Zhang et al. 2024), to representation-level interventions like Representation Misdirection (RMU) (Li et al. 2024) and the Erasure of Language

Memory (ELM) framework (Gandikota et al. 2026). While these techniques have shown encouraging results on standard benchmarks, evaluating whether unlearning has truly succeeded in the setting of Large Language Models (LLMs) remains fundamentally difficult (Shumailov et al. 2024). Unlike classification models, where the input-output pairs are well-defined, LLMs operate over ambiguous, open-ended inputs and outputs: the same piece of knowledge can be elicited through diverse phrasings, contextual cues, or multi-turn interactions. This ambiguity makes it challenging to definitively determine whether a model has genuinely forgotten a piece of information or has merely learned to suppress it under the narrow conditions tested by existing evaluations.

We identify two critical shortcomings in current unlearning benchmarks such as WMDP (Li et al. 2024), TOFU (Maini et al. 2024), and "Who's Harry Potter?" (Eldan and Russinovich 2023). First, these benchmarks evaluate knowledge retention using retain sets that are semantically distant from the unlearned content. For instance, WMDP uses the broad MMLU benchmark as its retain set, where only a small fraction of questions are topically related to microbiology and cybersecurity knowledge in the forget set. This means that severe degradation on closely related, safe knowledge, what we term *boundary* knowledge, can go entirely undetected while overall MMLU scores remain high. Second, existing benchmarks assess unlearning exclusively through direct, isolated queries on the forget set, without testing whether adversarial techniques can forcibly recover the supposedly erased information. Methods that merely suppress surface-level outputs can thus appear successful while leaving underlying knowledge structures intact and vulnerable to extraction via various jailbreaking attacks.

Although recent literature has highlighted some of these vulnerabilities (Rinberg et al. 2025; Lucki et al. 2024), the standard benchmarks used to evaluate unlearning algorithms still do not take this into account, which can lead to a false sense of progress. To address these gaps, we introduce *WMDP++*, an extension of the WMDP benchmark that incorporates two key evaluation dimensions: (1) a *Boundary* question set that probes performance on expert-level, safe questions in the near-distribution of the forget set, enabling detection of collateral damage that general benchmarks miss; and (2) a suite of adversarial robustness evaluations, including systematic in-context learning probes and

jailbreak attacks, that test whether unlearned information can be recovered under deliberate extraction attempts. Together, these additions provide a substantially more stringent and informative benchmark for evaluating unlearning in LLMs.

Background and Related Work

LLM Unlearning Machine unlearning refers to the targeted removal of specific training data influences, such as copyrighted material, private information, or hazardous knowledge, without the prohibitive cost of retraining the model from scratch (Liu et al. 2025; Ren et al. 2025). In the context of Large Language Models (LLMs) this can be complicated by the ambiguity of both inputs and outputs. The primary objective is to ensure safety and privacy compliance while preserving the model’s general utility and reasoning capabilities on unrelated tasks (Li et al. 2024). Methodologically, this is often achieved through optimization-based approaches such as Gradient Ascent (GA) or preference-based alignment. For instance, Negative Preference Optimization (NPO) and its simplified variant, SimNPO, adapt the principles of Direct Preference Optimization (DPO) to discourage the model from generating undesired information (Rafailov et al. 2023; Zhang et al. 2024; Fan et al. 2026). Advanced techniques such as Representation Misdirection (RMU) and approximate unlearning strategies break the conceptual link between prompts and sensitive outputs by steering intermediate activations toward random distributions (Li et al. 2024). Building upon these foundations, more recent research directions focus on fine-grained internal mechanics, such as the Erasing Conceptual Knowledge (ELM) framework or the use of Sparse Autoencoders to precisely suppress specific latent representations (Gandikota et al. 2026; Ashuach et al. 2025).

LLM Jailbreaking The vulnerability of Large Language Models (LLMs) to adversarial manipulation is often characterized through the lens of jailbreaking, a process where carefully crafted prompts bypass the safety guardrails and alignment mechanisms of the model to elicit prohibited content (Yi et al. 2024; Shen et al. 2023). Research in this domain has evolved from manual template-based attacks to automated optimization techniques. Notable among these is the Greedy Coordinate Gradient (GCG) method, which employs a gradient-based approach to find universal adversarial suffixes that can trigger harmful responses across multiple aligned models (Zou et al. 2023). Furthermore, black-box attacks like Crescendo (Russovich, Salem, and Eldan 2024) and Dialogue Injection Attack (DIA) (Meng et al. 2026) have been developed, utilizing an "attacker" and a "judge" models to refine and optimize semantic prompts in order to achieve a successful jailbreak without access to the model weights.

LLM Unlearning Benchmarks To evaluate the efficacy of unlearning algorithms, several benchmarks have been developed, typically focusing on a "forget set" (data to be removed) and a "retain set" (data to be preserved). Early efforts such as the "Who’s Harry Potter" benchmark (Eldan and Russovich 2023) utilize a GPT-4 evaluator to assess the removal of specific book series information while monitoring general quality through standard tasks like HellaSwag. More recent frameworks introduce diverse evaluation dimensions: TOFU

(Maini et al. 2024) focuses on unlearning synthetic datasets of fictional authors to measure precise information removal, while WMDP (Li et al. 2024) evaluates the redaction of hazardous expert-level knowledge in biology and cybersecurity with MMLU serving as the retain set. Despite these developments, recent frameworks such as MUSE (Shi et al. 2024) suggest that existing methods often struggle to balance utility preservation with privacy protection. Thaker et al. (2025) and Hu et al. (2025) showed that current benchmarks remain weak measures of overall progress, since they often fail to adequately measure the critical overlap between unlearning effectiveness and the preservation of required knowledge. Furthermore, Łucki et al. (2024) demonstrate that current benchmarks primarily verify the absence of knowledge under direct questioning, failing to account for more rigorous adversarial settings where supposedly removed capabilities can still be recovered through adaptive jailbreaking or fine tuning techniques.

WMDP++ Benchmark

Our goal is to create a more robust benchmark for unlearning in LLMs, to more faithfully evaluate current approaches. We adapt the WMDP benchmark to address two key limitations. First, the retain set is too dissimilar from the forget set, so we evaluate performance on closely related but safe concepts. Second, performance on the forget set is measured only via direct questioning, this is addressed via a variety of adversarial attempts to extract the unlearned information.

Boundary Concept Preservation

Current LLM unlearning benchmarks are typically structured around evaluating performance on a "forget set" (targeted data for removal) and a "retain set" (unrelated general knowledge for preservation). This paradigm focuses on minimizing accuracy or increasing perplexity on the forget set while maintaining standard benchmark scores on the retain set. While evaluating preservation on general knowledge is important, high scores can give a false sense of success. Due to the diversity of topics in the retain benchmarks, most questions differ substantially from the unlearned subject. A more challenging test is the "near distribution", the semantic and conceptual neighborhood surrounding the forget set. For example, if we aim to forget knowledge of how to create biological weapons, preserving safe microbiology knowledge is likely harder than preserving general history. Indeed, in practice, methods often exhibit "collateral damage" or "concept bleeding", where performance degrades on conceptually adjacent knowledge to the unlearn set, despite strong results on general retain benchmarks.

This critical gap is evident in the most widely used benchmarks: "Who’s Harry Potter?" (Eldan and Russovich 2023), TOFU (Maini et al. 2024), and WMDP (Li et al. 2024) benchmarks. In the WMDP benchmark (Li et al. 2024), the focus is on redacting hazardous expert-level biology and cybersecurity knowledge while preserving general MMLU scores. However, only a small percentage of MMLU questions are on related topics (biology or computer science), and an even smaller percentage is on closely related subtopics (microbiology and cybersecurity). As such, even if there is a significant

reduction in the performance of the model on closely related subtopics, it will have a minimal impact on the overall MMLU score. We note that some unlearning methods, such as CRISP (Ashuach et al. 2025), report MMLU score on the specific topic (e.g., biology), but as we will show, this is still too general to capture the boundary performance. Furthermore, this is not part of the standard benchmark, so this is not ubiquitous.

To address these limitations, we have developed a Boundary queries dataset, a new evaluation benchmark that is specifically designed to probe the near-distribution of the WMDP hazardous questions. This dataset consists of high-quality, expert-level questions spanning the biology and cybersecurity domains that are conceptually adjacent to the "forget" knowledge but are fundamentally safe for public release. Unlike standard retain sets like MMLU, which often cover broad, general-purpose knowledge, our Boundary dataset targets the "gray area" of expertise-challenging the model to maintain its sophisticated scientific reasoning and technical proficiency in legitimate research areas while its hazardous capabilities are removed. In addition, this differs from recent work such as BLUR (Hu et al. 2025), which concatenates separate forget and retain queries into a single prompt. Our Boundary questions are semantically near-distribution to hazardous categories but strictly safe, isolating the effect of conceptual adjacency alone. These questions were generated through an iterative process using the GPT-5.2 model: first, we extracted 25 core categories from the original hazardous WMDP questions to identify critical expertise areas; then, we iteratively synthesized 100 safe, near-distribution questions per domain (4 per category) that mirror the technical depth and linguistic patterns of the hazardous set but remain strictly non-malicious. We note that while the boundary dataset is relatively small, with only 100 questions, it is used solely for evaluation rather than training or fine-tuning. As such, it is sufficient for assessing whether there was a significant degradation in performance.

While the questions were curated and evaluated by an LLM, we used a human expert to evaluate the questions in the cybersecurity benchmark. These were indeed validated as relevant, correct, and safe. Unfortunately, testing the validity of the biology domain questions was a much more demanding task, as it required a significant amount of time from highly specialized experts. We believe that the human evaluation in the cybersecurity domain shows that our pipeline and the stronger LLM-judge are able to generate proper questions and answers. Moreover, the strong LLM-judge model is deployed with guardrails to prevent it from generating hazardous questions (at least not without jailbreaking attempts) further ensuring the questions are safe. Given the safety of the questions in the boundary benchmark, a significant drop in accuracy shows deviation from the base model on safe questions, which is undesirable. The detailed methodology for the question generation and evaluation is provided in the Appendix.

Robust Unlearning Evaluation

Traditional unlearning metrics offer only a narrow view of model safety, as they evaluate performance through direct,

isolated queries on the forget set. While such evaluations can verify whether a model suppresses targeted knowledge during standard interactions, they overlook alternative adversarial settings in which deliberate techniques are used to elicit the supposedly forgotten information. If a model has truly unlearned something, no method should be capable of recovering it. When one does, it reveals that the information has been suppressed rather than genuinely forgotten.

This limitation is pervasive across dominant unlearning benchmarks such as "Who's Harry Potter?" (Eldan and Russinovich 2023), WMDP (Li et al. 2024), and TOFU (Maini et al. 2024), which typically rely on straightforward multiple-choice or direct question-answering pairs to evaluate knowledge removal. By failing to incorporate adversarial components or complex prompting into their standard protocols, these benchmarks may significantly overestimate unlearning efficacy. Methods that merely suppress the probability of direct answers can appear successful while leaving underlying knowledge structures vulnerable to retrieval via indirect techniques.

To address these vulnerabilities, we introduce a more comprehensive evaluation framework that incorporates adversarial robustness as a primary metric. In our evaluation, we subject the unlearned model to several adversarial tests to ensure that the unlearning is robust. First, we employ In-Context Learning (ICL) by providing context information with various degrees of relevance. We experiment with paragraphs sourced from the forget set, retain set, or Wikipedia. Furthermore, we experiment with utilizing QA pairs from the relevant subset of MMLU, our related near-distribution questions, or the forget test set, as detailed in the Appendix. These contexts and QA pairs are prepended to the current question to "nudge" the model toward its original knowledge. This allows us to explore how the model performs on the unlearned task, as the additional knowledge becomes closer and closer to the forget set. In the extreme case where we add questions from the forget test set, it simulates the situation where a person with some knowledge in the hazardous domain is trying to use the model to further extend his or her knowledge.

Furthermore, we evaluated our approach against several jailbreaking methodologies. First, we employ a white-box attack using an enhanced Greedy Coordinate Gradient (GCG) method (Łucki et al. 2024). We also evaluated our approach against black-box methodologies, specifically Crescendo (Russovich, Salem, and Eldan 2024) and Dialogue Injection Attack (DIA) (Meng et al. 2026), to rigorously test the robustness of our framework under different adversarial settings. While we also attempted to use Prompt Automatic Iterative Refinement (PAIR) (Chao et al. 2025), this method proved unreliable for our specific objectives; the generated prefixes often inadvertently leaked clues or partial answers about the target queries. Consequently, PAIR was not utilized for the final evaluation, as further detailed in the Appendix. By prepending these adversarial prefixes and "nudge" contexts before target queries, we provide a significantly more robust measure of whether supposedly forgotten knowledge has been truly removed or merely suppressed.

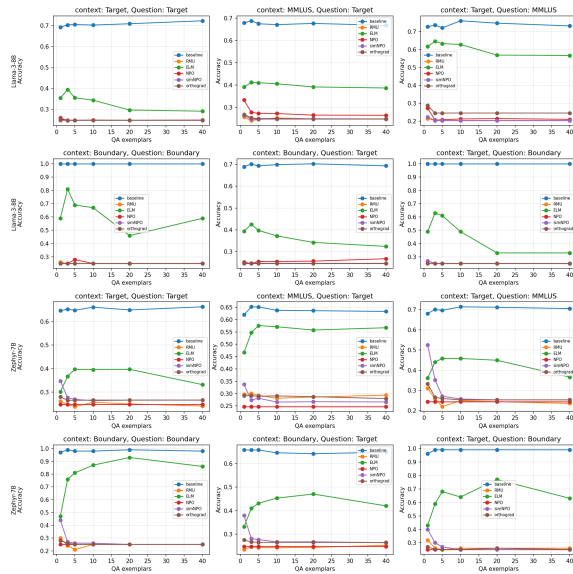


Figure 1: Biology domain: This context comprises QA pairs from the Target, MMLUS, or Boundary sets, matching questions from the respective domains. The first two rows present results for the Llama-3-8B model, while the final two rows display the results for the Zephyr-7B model.

Experiments

We will describe in this section our experimental setup, and show that it allows us to better evaluate existing unlearning methods, giving a more reliable measure of success.

Experimental Setup

Models We perform our experiments on two open-source LLMs, we use Llama-3-8B (Grattafiori et al. 2024), and the Zephyr-7B Beta (Tunstall et al. 2023) models.

Unlearning Methods We evaluate five approaches: , Random Misdirection for Unlearning (RMU) (Li et al. 2024), Erasure of Language Memory (ELM) (Gandikota et al. 2026), Negative preference optimization (NPO) (Zhang et al. 2024), simNPO (Fan et al. 2026) and OrthoGrad (Shamsian et al. 2025). These methods and the hyper-parameters used for each method are described in more detail in the "Existing Unlearning Techniques" Appendix.

New Evaluation Metrics Our evaluation framework systematically assesses the efficacy and robustness of various unlearning methods through several dimensions. On top of the standard WMDP metrics, target accuracy to measure unlearning and MMLU to measure general knowledge retention, we add two new metrics. We evaluate the unlearning with the Max-JB metric, which gives the highest accuracy from a diverse set of jail-breaking methods (described in the "Robust Unlearning Evaluation" section). This is a much more challenging metric as it does not allow the unlearned data to just be suppressed, but tests more thoroughly that it has been removed. Finally, to evaluate knowledge retention, we evaluate accuracy on our generated list of Boundary

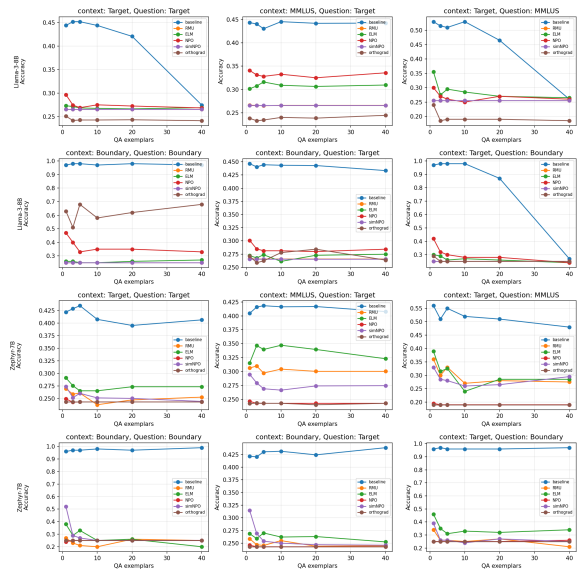


Figure 2: Cybersecurity domain: This context comprises QA pairs from the Target, MMLUS, or Boundary sets, matching questions from the respective domains. The first two rows present results for the Llama-3-8B model, while the final two rows display the results for the Zephyr-7B model.

problems that are benign but are much more similar to the unlearned knowledge.

Main Results

Inadequacy of Target Metrics in Distinguishing Model Erasure The primary results across both domains (Tables 1 and 2) reveal that the conventional target forget set accuracy (Target) fails to provide a meaningful distinction between different unlearning methods, offering a deceptive signal of model safety. For instance, in the Biology domain (Table 1) on Llama-3-8B, RMU and simNPO achieve nearly identical low target accuracies of 24.98% and 26.08%, respectively. However, under adversarial pressure, RMU maintains robust erasure with a Max-JB score of only 28.12%, whereas simNPO severely degrades, surging to 60.09% accuracy. In general, RMU exhibits significantly superior robust unlearning results, which is not apparent in the target metric. The stark discrepancies between Target and Max-JB show that standard target metrics fail to properly rank or differentiate unlearning approaches, obscuring whether a method induces true weight-level erasure or merely superficial suppression.

Inadequacy of General Benchmarks for Near-Distribution Knowledge Standard evaluation metrics fail to detect the severe performance degradation that occurs in benign knowledge closely related to the target forget data. General benchmarks like MMLU and its high school or college domain-specific subset (MMLUS) appear largely stable, masking the sharp degradation of safe expertise. However, the performance on near-distribution questions (Boundary) reveals that unlearning severely damages closely related benign knowledge. For example, Zephyr-7B with

Model	Method	Target ↓	Max-JB ↓	MMLU ↑	MMLUS ↑	Boundary ↑
Llama-3-8B	Base	0.6913	-	0.6008	0.7467	1.0000
	RMU	0.2498	0.2812	0.4022	0.3789	0.2600
	ELM	0.3323	0.4910	0.5393	0.6630	0.4800
	NPO	0.2946	0.5420	0.4901	0.5419	0.3600
	simNPO	0.2608	0.6009	0.3714	0.4185	0.2900
	OrthoGrad	0.2742	0.6245	0.5379	0.6211	0.3800
Zephyr-7B	Base	0.6591	-	0.5869	0.7203	0.9800
	RMU	0.3166	0.3637	0.5761	0.6916	0.4700
	ELM	0.2663	0.5405	0.5279	0.4626	0.2800
	NPO	0.2529	0.5169	0.4227	0.3744	0.2600
	simNPO	0.4564	0.4140	0.4740	0.6013	0.6700
	OrthoGrad	0.3071	0.4737	0.5272	0.5771	0.3600

Table 1: WMDP Biology Domain Combined Results: Performance across all evaluated metrics including the forget target set (Target), worst-case jailbreak extraction (Max-JB), general MMLU biology (MMLU), high school and college biology subset (MMLUS), and near distribution questions (Boundary). The best result per metric is presented in bold.

Model	Method	Target ↓	Max-JB ↓	MMLU ↑	MMLUS ↑	Boundary ↑
Llama-3-8B	Base	0.4444	-	0.6008	0.5550	0.9700
	RMU	0.2657	0.2657	0.2302	0.2550	0.2500
	ELM	0.3015	0.3281	0.5393	0.5150	0.3800
	NPO	0.3619	0.4172	0.4901	0.4250	0.7700
	simNPO	0.2758	0.3865	0.3714	0.2900	0.3500
	OrthoGrad	0.2763	0.3996	0.4111	0.2900	0.5700
Zephyr-7B	Base	0.4177	-	0.5869	0.5650	0.9600
	RMU	0.3040	0.3181	0.5761	0.5200	0.4600
	ELM	0.2934	0.3488	0.5279	0.4800	0.3500
	NPO	0.3251	0.3684	0.4227	0.3800	0.5600
	simNPO	0.3402	0.3347	0.4740	0.5000	0.6200
	OrthoGrad	0.2496	0.3447	0.4411	0.2950	0.3000

Table 2: WMDP Cybersecurity Domain Combined Results: Performance across all evaluated metrics including the forget target set (Target), worst-case jailbreak extraction (Max-JB), general MMLU cybersecurity (MMLU), high school and college biology MMLU subset (MMLUS), and near distribution questions (Boundary). The best result per metric is presented in bold.

RMU in the Biology domain (Table 1) shows minimal drops of roughly 1% on MMLU and 3% on MMLUS, yet its accuracy on Boundary questions plummets by 51%. Similarly, Llama-3-8B with ELM in Cybersecurity (Table 2) exhibits a drop of 6% on MMLU and 4% on MMLUS, contrasting sharply with a 59% collapse on the Boundary distribution. This demonstrates that standard MMLU benchmarks, even when narrowed to domain-specific subsets, are too coarse to capture concept bleeding into near-distribution knowledge.

Detailed Results

In addition to our novel metric evaluations, we conduct extensive probing experiments to thoroughly analyze the knowledge persistence under varied extraction techniques. Specifically, we test model resilience against contextual nudging by examining performance when evaluation queries are preceded by few-shot question-answer (QA) demonstrations. Furthermore, we dissect the relative efficacy of distinct adversarial jailbreak paradigms, comparing white-box opti-

mization techniques against black-box and multi-turn attack strategies to evaluate the underlying robustness of each unlearning method. These jailbreaking methods and the hyperparameters used for each method are described in more detail in the "Jailbreak Methods Evaluation" Appendix.

Sensitivity to In-Context QA Demonstrations In this section, we evaluate the robustness of unlearned models under contextual nudging by prepending relevant question-answer (QA) pairs prior to querying the model. Our empirical findings indicate that providing prepended QA context generally fails to facilitate knowledge recovery across most evaluated setups, often leaving performance near random-chance levels or even causing further degradation. As shown in Tables 3 and 4, methods such as RMU, NPO, simNPO, and OrthoGrad consistently decay or remain stagnant near 24%–28% accuracy when provided with few-shot QA context. However, ELM in the Biology domain emerges as a notable exception, demonstrating a distinct vulnerability where contextual steering leads to a substantial recovery of sup-

Model	Method	Target ↓	Target-Target ↓	Boundary-Target ↓	MMLUS-Target ↓
Llama-3	Base	0.6913	0.7093	0.7023	0.6764
	RMU	0.2498	0.2467	0.2467	0.2474
	ELM	0.3323	0.2962	0.3425	0.3912
	NPO	0.2946	0.2474	0.2569	0.2647
	simNPO	0.2608	0.2467	0.2467	0.2467
	OrthoGrad	0.2742	0.2474	0.2474	0.2482
Zephyr	Base	0.6591	0.6496	0.6418	0.6363
	RMU	0.3166	0.2490	0.2443	0.2859
	ELM	0.2663	0.3967	0.4705	0.5577
	NPO	0.2529	0.2474	0.2474	0.2474
	simNPO	0.4564	0.2655	0.2679	0.2671
	OrthoGrad	0.3071	0.2655	0.2655	0.2875

Table 3: Accuracy on Target biology questions under different evaluation setups. The first column denotes zero-shot accuracy without additional context. The subsequent columns show performance when the evaluation questions are preceded by few-shot QA contexts derived from various sources (Target, Boundary, or MMLUS). The best result per metric is presented in bold.

Model	Method	Boundary ↑	Target-Boundary ↑	Boundary-Boundary ↑
Llama-3	Base	1.0000	1.0000	1.0000
	RMU	0.2600	0.2500	0.2500
	ELM	0.4800	0.3300	0.4600
	NPO	0.3600	0.2500	0.2500
	simNPO	0.2900	0.2500	0.2500
	OrthoGrad	0.3800	0.2500	0.2500
Zephyr	Base	0.9800	0.9900	0.9900
	RMU	0.4700	0.2600	0.2500
	ELM	0.2800	0.7700	0.9300
	NPO	0.2600	0.2500	0.2500
	simNPO	0.6700	0.2500	0.2500
	OrthoGrad	0.3600	0.2600	0.2500

Table 4: Accuracy on Boundary biology questions under different evaluation setups. The first column reports zero-shot accuracy without additional context. The subsequent columns present performance when preceding the evaluation questions with few-shot QA contexts sourced from Target or Boundary. The best result per metric is presented in bold.

pressed information. As illustrated in Figure 1, while all other unlearning methods remain flat regardless of the context length, ELM generally exhibits improved performance as the number of prepended QA exemplars increases, before experiencing a minor drop at higher exemplar counts.

Efficacy of Jailbreak Strategies on Knowledge Extraction

The evaluation of adversarial jailbreak attacks across both domains (Tables 5 and 6) demonstrates the varying capability of different attack paradigms to recover suppressed knowledge. In the vast majority of evaluated configurations, white-box optimization via the enhanced GCG attack significantly outperforms black-box and multi-turn methods like DIA and Crescendo, revealing that models are relatively safe without full access. This white-box dominance is clearly illustrated in Llama-3-8B within the Biology domain (Table 5) under OrthoGrad, where the benign target accuracy of 27.42% surges to 62.45% under GCG, while black-box attacks like DIA (24.74%) and Crescendo (24.74%) fail to induce any recovery. A similar pattern appears in Cybersecurity (Table 6)

for Llama-3-8B under simNPO, where GCG elevates performance from 27.58% to 38.65%, whereas DIA (26.57%) and Crescendo (26.57%) remain completely ineffective. However, this trend is not universal, as notable exceptions exist where black-box techniques prove superior. Specifically, Zephyr-7B unlearned with ELM in Biology (Table 5) exhibits an inverse vulnerability: while GCG recovers performance to 49.25%, the multi-turn Crescendo attack achieves an even higher recovery of 54.05% from a 26.63% benign baseline. Likewise, under RMU on Zephyr-7B in Cybersecurity (Table 6), DIA achieves a slightly higher extraction accuracy (31.81%) than GCG (31.00%). These findings indicate that while models are generally far more vulnerable to white-box gradient extraction, specific unlearning paradigms remain uniquely susceptible to contextual and black-box jailbreak strategies.

Model	Method	Target ↓	GCG ↓	DIA ↓	Crescendo ↓
Llama-3	Base	0.6913	-	-	-
	RMU	0.2498	0.2812	0.2569	0.2482
	ELM	0.3323	0.4910	0.2592	0.2498
	NPO	0.2946	0.5420	0.2529	0.2529
	simNPO	0.2608	0.6009	0.2467	0.2467
	OrthoGrad	0.2742	0.6245	0.2474	0.2474
Zephyr	Base	0.6591	-	-	-
	RMU	0.3166	0.3637	0.3087	0.2608
	ELM	0.2663	0.4925	0.4580	0.5405
	NPO	0.2529	0.5169	0.2718	0.2474
	simNPO	0.4564	0.4140	0.2781	0.3354
	OrthoGrad	0.3071	0.4737	0.2844	0.2962

Table 5: Model accuracy on Target biology questions under various jailbreak attacks. The first column (Before) reports baseline accuracy under standard benign prompting. Subsequent columns show performance under adversarial attacks using GCG, DIA, and Crescendo. The best result per metric is presented in bold.

Model	Method	Target ↓	GCG ↓	DIA ↓	Crescendo ↓
Llama-3	Base	0.4444	-	-	-
	RMU	0.2657	0.2647	0.2657	0.2657
	ELM	0.3015	0.3281	0.2743	0.2672
	NPO	0.3619	0.4172	0.2783	0.2687
	simNPO	0.2758	0.3865	0.2657	0.2657
	OrthoGrad	0.2763	0.3996	0.3281	0.2406
Zephyr	Base	0.4177	-	-	-
	RMU	0.3040	0.3100	0.3181	0.2677
	ELM	0.2934	0.3488	0.2964	0.3105
	NPO	0.3251	0.3684	0.3578	0.2693
	simNPO	0.3402	0.3347	0.2738	0.2874
	OrthoGrad	0.2496	0.3447	0.2360	0.2431

Table 6: Model accuracy on Target cybersecurity questions under various jailbreak attacks. The first column (Before) reports baseline accuracy under standard benign prompting. Subsequent columns show performance under adversarial attacks using GCG, DIA, and Crescendo. The best result per metric is presented in bold.

Conclusion

In this work, we introduced WMDP++, an enhanced evaluation framework designed to address critical blind spots in current large language model unlearning benchmarks. By incorporating our Boundary dataset, we demonstrate that existing unlearning algorithms often cause severe, undetected collateral damage to safe, conceptually adjacent near-distribution knowledge. Furthermore, our robustness evaluations under adversarial jailbreaks and in-context steering reveal that many state-of-the-art methods merely suppress surface-level outputs rather than genuinely erasing latent knowledge. Ultimately, WMDP++ establishes a more realistic standard for LLM safety, highlighting the urgent need for future unlearning methodologies to guarantee both robust erasure and precise boundary preservation.

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